

LR-CSW-51T2 + LR-NCE-A0 — ASIC Design Package

Date: 2026-04-19 • **Distribution:** Investor data-room **Process node:** TSMC N3 • **Owner:** LightRail AI — Silicon Engineering

1. System overview

LightRail AI's silicon stack comprises two custom ASICs and a co-packaged optics (CPO) module, integrated on a single CoWoS-L composite for the switch side and an FCBGA for the compute side.

Part	Function	Process	Package	Power
LR-CSW-51T2	51.2 Tbps CPO switch ASIC	TSMC N3	CoWoS-L composite (80×80 mm with 16 CPO engines on interposer)	350 W
LR-NCE-A0	Neural compute engine ASIC	TSMC N3	FCBGA 65×65 mm	600 W
LR-CPO-3T2-A0	3.2 Tbps CPO engine (×16 per switch)	TSMC N6 + TFLN PIC	On CoWoS-L interposer	5 W ea

2. LR-CSW-51T2 — switch ASIC

2.1 Block hierarchy

lr_csw_top	
├ 16 × lr_csw_quad	– quadrant (16 lanes × 200 G PAM4 + parser + sched)
│ └ 16 × lr_serdes_lane	– 224 G PAM4 SerDes hard-IP wrapper
│ └ lr_csw_parser	– Ethernet RX parser, 9 KB MTU
│ └ lr_csw_sched	– DRR scheduler / egress queue
├ lr_csw_fabric	– 80×80 crossbar (16 ingress × 16 egress, fair round-robin)
├ 32 × lr_ucie_bridge	– UCle-D2D bridge to on-interposer CPO engines (4 GHz)
├ lr_reg_bank	– APB CSR + IRQ aggregation (BMC sideband)
├ lr_jtag_tap	– IEEE 1149.1 / 1149.6 TAP
└ common library	– lr_reset_sync, lr_cdc_2ff, lr_async_fifo, lr_arb_rr

2.2 Clock domains (10)

Domain	Frequency	Purpose
clk_core	2.0 GHz	crossbar fabric + parser/sched
clk_mem	1.5 GHz	HBM scratchpad logic
clk_serdes_tx[0:15]	1.6 GHz × 16	per-quadrant TX
clk_serdes_rx[0:15]	1.6 GHz × 16	per-quadrant RX
clk_ucie_tx / clk_ucie_rx	4.0 GHz	UCle D2D
clk_pcie_aux	250 MHz	PCIe Gen 6 aux
clk_jtag	50 MHz	TAP
clk_refclk	156.25 MHz	reference

Domain	Frequency	Purpose
clk_aux	100 MHz	APB CSR + BMC

All cross-domain crossings use lr_cdc_2ff or lr_async_fifo; sign-off in lint/questa_cdc/cdc.violations.rpt — 0 errors.

2.3 Reset domains (4)

Reset	Scope	Strategy
por_rst_n	full chip	async-assert / sync-deassert, lr_reset_sync per clock domain
warm_rst_n	datapath only	clears MAC tables, keeps CSR
soft_rst_n	per-quadrant	re-runs SerDes training
jtag_rst_n	TAP only	independent of system

Sign-off in lint/questa_rdc/rdc.violations.rpt — 0 errors.

2.4 Power domains (6, UPF 3.1)

Domain	Voltage	Switching	Retention
PD_TOP	0.85 V	always-on	n/a
PD_AUX	0.85 V	always-on	CSR shadow retained
PD_CORE	0.85 V (VAVS)	DVFS via PMBus	scoreboard counters
PD_UCIE	0.85 V	switchable	none
PD_SERDES	0.85 V analog	switchable	none
PD_MEM	1.20 V	switchable	none

2.5 Datapath bandwidth

256 × 224 G PAM4 (CP0 side) = 57.3 Tbps raw → 51.2 Tbps after FEC + protocol overhead

3. LR-NCE-A0 — compute ASIC

3.1 Block hierarchy

lr_nce_top	
├ 4 × lr_nce_cluster	– 64 systolic lanes each, FP8 + INT8 + FP16
├ 4 × lr_hbm4_ctl	– 1024-bit HBM4 controllers (6.4 Tbps aggregate)
├ lr_pcie6_x16	– PCIe Gen 6 ×16 host link
├ 4 × lr_ucie_bridge	– UCIE D2D to switch ASIC
├ lr_reg_bank	– APB CSR
└ common library	– shared with LR-CSW

3.2 Memory bandwidth

4 × HBM4 stacks × 1024-bit × 1.6 GT/s = 6.4 Tbps

4. Verification methodology

Refer to Tape-Out_Readiness_Signoff.md §3 for the full nine-discipline closure table. Highlights:

- UVM testbench with `lr_csw_env_pkg` (driver, monitor, scoreboard, coverage subscriber) and `lr_csw_smoke_test`.
- Makefile targets: compile, smoke, regress (4 000 jobs), coverage (vcover merge + report).
- 11-rule lint sign-off (Questa Lint) with synthesisable-subset enforcement.
- Questa CDC + RDC with metastability injection (1 000 iterations).
- PrimeTime + Tempus STA across slow / typical / fast corners.
- Voltus EM/IR static + dynamic.
- Calibre DRC / LVS / antenna against TSMC N3 PDK.
- Conformal LEC RTL ↔ post-PnR netlist.

5. FPGA prototype (AMD Versal Premium VP2802)

Architecture, IP integration, and partitioning challenges documented in `docs/LR-DV_Digital_Design_Verification_FPGA_Plan.md`. Single-board proof on a VP2802 with a 4-board partition path to full 51.2 Tbps line-rate emulation. ILA + VIO + JTAG-to-AXI instrumentation for live debug.

6. Release contents

```
LR-DV/
├─ rtl/{common,lr_csw,lr_nce}/*.sv
├─ tb/{uvm,sva,Makefile}
├─ syn/{genus,dc}/run_*.tcl, sdc/lr_csw.sdc, upf/lr_csw.upf
├─ pnr/innovus/run_innovus.tcl
├─ signoff/{sta,power,formal,drc_lvs}/run_*.tcl
├─ lint/{questa_lint,questa_cdc,questa_rdc}/lr_*.do
├─ fpga/vivado/{tcl,xdc,ip}/
├─ docs/ (this package + tape-out checklist + sign-off)
└─ reports/ (clean sign-off reports – STA, EM/IR, lint, CDC, RDC, formal, DRC)
```

7. Sign-off

See Tape-Out_Readiness_Signoff.md §6 — 10-row approval table dated 2026-04-19, all LightRail AI engineering and programme functions approved for foundry data-room release.